Letter Sequences & Alphabet Code (Set B)

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Example 1 — Letter Sequence: What comes next in this sequence? B, D, F, H, __

Answer: J — each letter skips one letter in the alphabet (+2 each time): B(2), D(4), F(6), H(8), J(10).

Example 2 — Position Code: If A=1, B=2, C=3 and so on, what number represents the letter K?

Answer: 11 — K is the 11th letter of the alphabet.

Example 3 — Shift Code: In a code where every letter is shifted forward by 3 places (A→D, B→E, C→F...), what does the letter M become?

Answer: P — M is shifted 3 places forward: M, N, O, P.

Spotting the Pattern (15 questions)

1. What comes next? A, C, E, G, __
2. What comes next? Z, Y, X, W, __
3. If A=1, B=2, C=3..., what number stands for the letter F? A) 5 B) 6 C) 7 D) 8 E) 9
4. Select ALL that apply: which of these letter pairs are exactly 2 letters apart in the alphabet (e.g. B and D)? A) C and E B) G and I C) M and O D) P and R E) T and U
5. What comes next in this sequence? B, E, H, K, __
6. In a code where A=1, B=2, C=3..., which letter stands for the number 15? A) M B) N C) O D) L E) K
7. What comes next? Y, W, U, S, __
8. Fill in the missing letter: A, _, E, G, I (skipping one letter each time)
9. If every letter is shifted forward by 1 place (A→B, B→C, C→D...), what does D become? A) C B) E C) F D) G E) B
10. Select ALL that apply: in the sequence C, F, I, L, O, which of these statements are true? A) The gap between each letter is always 3. B) The next letter after O would be R. C) The sequence goes backwards through the alphabet. D) The first letter is C. E) The letter after L is M.
11. What comes next? D, F, H, J, __
12. In the alphabet-position code (A=1, B=2, ...), what number represents the letter T? A) 18 B) 19 C) 20 D) 21 E) 22
13. What comes next? Q, O, M, K, __
14. If letters are shifted BACK by 2 places (C→A, D→B, E→C...), what does H become? A) E B) F C) G D) I E) J
15. Select ALL that apply: which of these are letters that come AFTER M in the alphabet? A) P B) J C) R D) L E) W

Applying the Pattern (15 questions)

1. What comes next in this sequence? B, D, G, I, L, __ (the gap alternates +2, +3, +2, +3...)

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2. A code shifts every letter forward by 4 places (A→E, B→F…). What does the word "CAB" become?

3. What comes next? A, D, H, M, __ (the gap increases by 1 each time: +3, +4, +5…)

4. In a code, A=1, B=2, C=3… Two numbers are added together and converted back to a letter: if the numbers are 7 and 8, which letter results? A) N B) O C) P D) M E) Q

5. Select ALL that apply: which sequences below correctly continue with a GAP of 4 between each letter? A) A, E, I, M, Q B) B, F, J, N, R C) C, F, I, L, O D) D, H, L, P, T E) A, D, G, J, M

6. What comes next? Z, W, T, Q, __ (the gap is always -3)

7. A code shifts every letter BACK by 3 places (D→A, E→B, F→C…). What does the word "GIF" decode to?

8. What comes next in this sequence, which alternates two interleaved patterns? A, Z, B, Y, C, __

9. In the position code (A=1…Z=26), what letter corresponds to the number you get from 26 − 6? A) S B) T C) U D) V E) W

10. Select ALL that apply: in a +5 shift code (A→F, B→G, C→H…), which of these letter pairs are correctly matched? A) A→F B) C→H C) J→O D) M→R E) Z→E (wrapping to the start of the alphabet)

11. What comes next? C, F, J, O, __ (the gap increases by 1 each time: +3, +4, +5…)

12. A code uses the rule "position number + 3, then convert back to a letter". What does the letter W become? (wrap round to A after Z if needed)

13. What comes next? L, I, F, C, __

14. Select ALL that apply: which of these statements about the sequence E, H, K, N, Q are correct? A) Each gap is +3. B) The sequence is decreasing. C) The next letter after Q is T. D) The first letter is E. E) The gap between K and N is 2.

15. In a shift code where every letter moves forward by 7 places, what does the letter S become? (wrap round to the start of the alphabet if you go past Z)

Codes in Context (15 questions)

1. What comes next in this sequence, and explain the rule: B, C, E, H, L, __ (hint: look at the GAPS between letters)

2. A code shifts every letter forward by 5 places, wrapping round after Z. Decode the word "YZ".

3. In a position code (A=1…Z=26), the "code number" of a word is the SUM of its letters' positions. What is the code number of the word "CAB"?

4. Select ALL that apply: in the sequence Z, X, U, Q, __, which of the following are true about the pattern? A) The gaps are -2, -3, -4… B) The next letter is N. C) The sequence increases through the alphabet. D) The gap between Z and X is 2. E) The gap between X and U is 3.

5. A code works like this: each letter's position number is doubled, then converted back to a letter (wrapping round after Z if needed). What does the letter D become?

6. What comes next in this sequence, which interleaves two separate alphabet sequences going in opposite directions? A, Z, C, X, E, __, __

7. In a shift code, the letter F becomes K. Using the same shift, what would the letter Y become (wrap round after Z)?

8. A puzzle gives you this pair as a worked example of a code: PEN → SHQ. Work out the shift used, then decode the word "VQI" using the same shift.

9. Select ALL that apply: which of these are valid ways to describe the sequence D, G, J, M, P? A) Each letter's position number increases by 3. B) The letters are every 3rd letter of the alphabet starting from D. C) The

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sequence could also be written using position numbers as 4, 7, 10, 13, 16. D) The next letter after P would be R. E) The gap is consistent throughout the sequence.

10. A code reverses the alphabet (A ↔ Z, B ↔ Y, C ↔ X…). What does the word "CAB" become under this code?

11. What comes next in this sequence, and what is the underlying rule? A, B, D, G, K, __ (hint: look at how much the gap increases each time)

12. In a position-number code, a word's code is found by taking each letter's position and subtracting 1 (so A=0, B=1, C=2…). What 3-letter word has the code 3, 4, 8?

13. A code shifts vowels (A, E, I, O, U) forward by 2 places but leaves all consonants unchanged. Decode the effect on the word "CAT".

14. Select ALL that apply: a student claims the sequence W, T, Q, N, K continues correctly with H as the next letter. Which of these statements support or refute this? A) The gap throughout the sequence is -3. B) W is the 23rd letter, so the next term should be 23 − (3×4) = 11, which is K, not H. C) K is the 11th letter. D) H is the correct next letter. E) The pattern is consistent, so H must be next.

15. A two-step code first shifts every letter forward by 2 places, then reverses the order of the letters in the word. Apply this to the word "BAT" and give the final 3-letter result.

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